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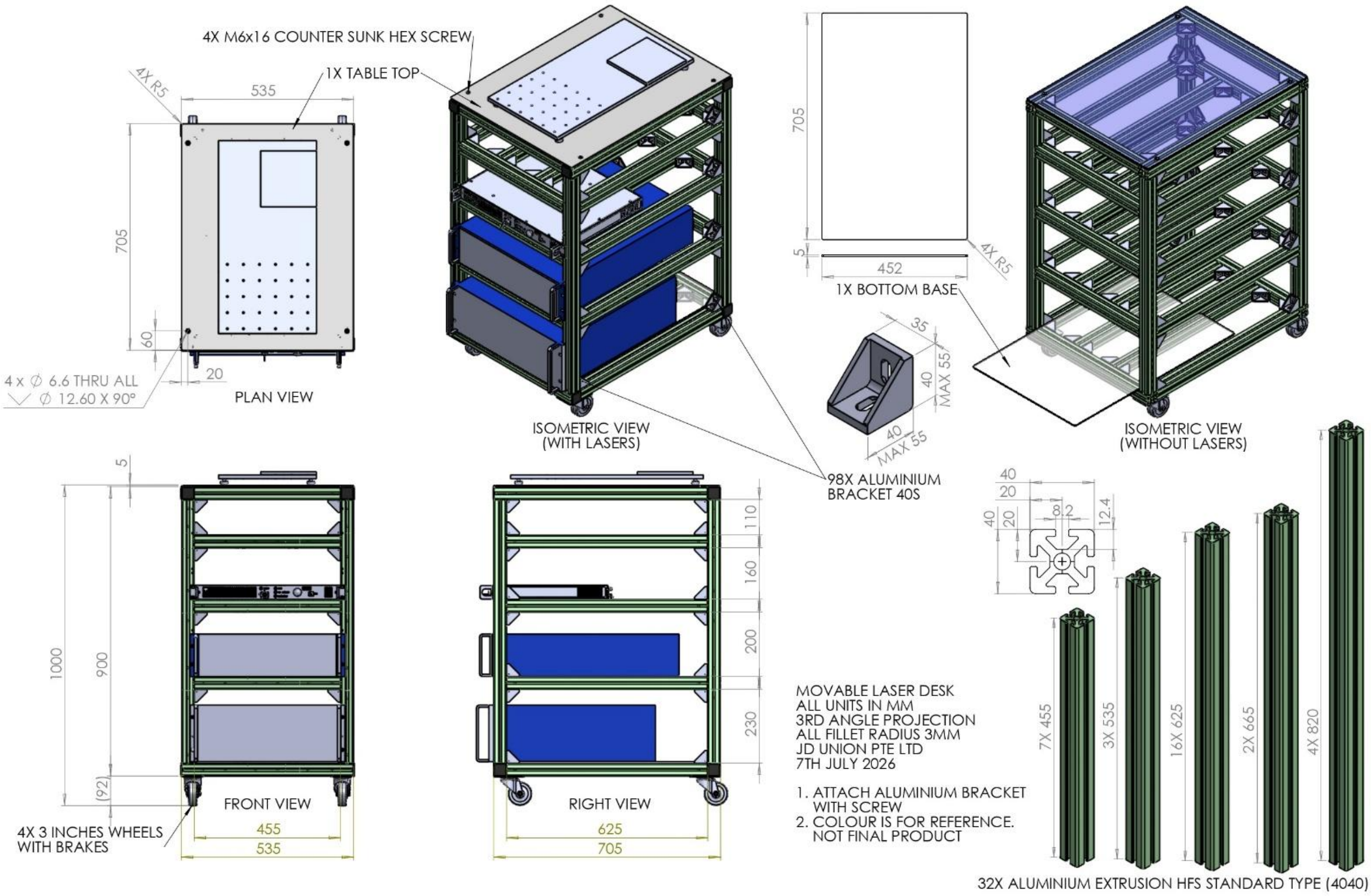
HARDWARE PORTFOLIO HIGHLIGHTS

Figure 1. Mobile Laser Test Workstation (JD Union, 2026)

Constraints:	Vibration corrupts sub-micron laser measurements before the optics even engage.
Design Structure:	To isolate dynamic vibrations across a >250kg footprint, I engineered the 5-tier frame using 4040 aluminium extrusion, balancing rapid CNC turnaround and reconfigurability with strict geometric squareness.
Execution:	Generated DFM-compliant SolidWorks drawings for outsourced fabrication, followed by full physical commissioning of the IPG fiber laser, chiller, and power supply.



Explore the interactive 3D CAD model and complete project documentation at ***Clement.Builds***

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Figure 2. EOS M 290 Custom DAQ Sensor Integration (A*STAR ARTC, 2023)

Constraints:	Extracting live process telemetry from a commercial LPBF printer required routing external DAQ hardware without compromising the chamber's strict <0.1% oxygen atmospheric seal or introducing spark risks.
Design Structure:	Engineered a non-intrusive sensor integration utilizing a custom 3D-printed PCB holder, terminal blocks, and a 200mA fast-acting fuse to bridge the internal powder chamber with external data systems.
Execution:	Safely routed 12 distinct signal wires from internal strain gauges and acoustic emission sensors through custom-machined 13mm ports, establishing the physical foundation for the Python data pipeline.

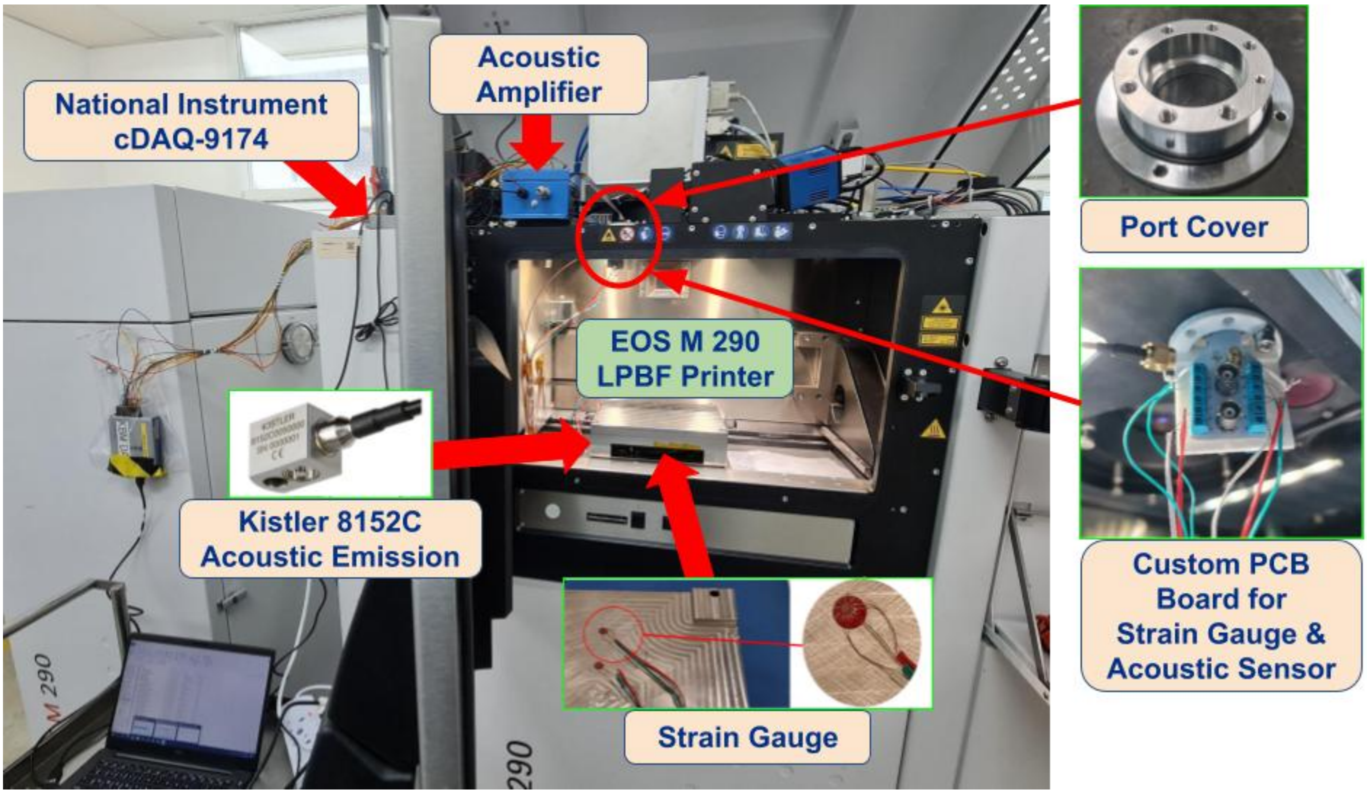

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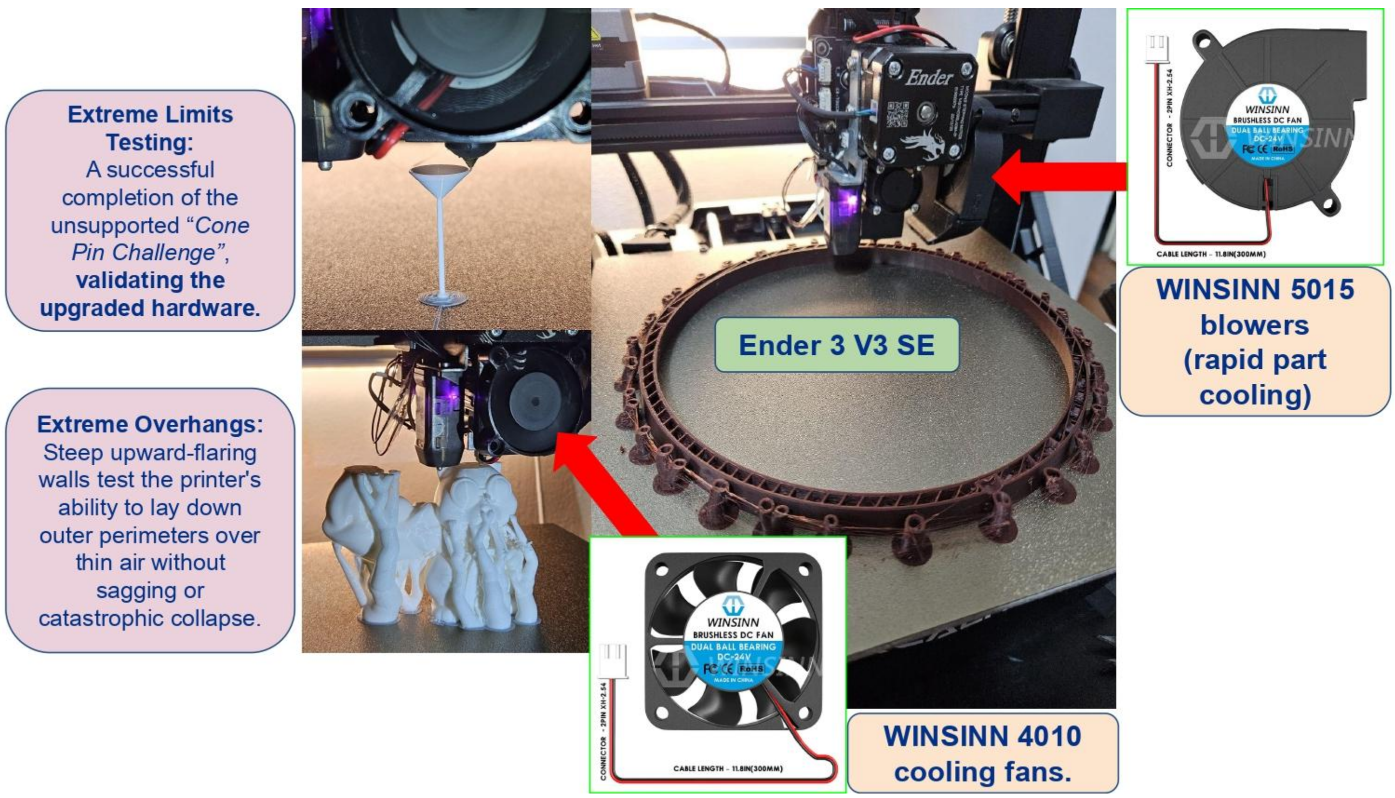
Figure 3. 3D Printer Thermal & Hardware Optimization (Ender 3 V3 SE, 2024)

Constraints:	Inconsistent thermal dissipation at the hot-end caused severe part warpage, degrading print reliability and the overall hardware user experience.
Design Structure:	To optimize laminar airflow, I continuously prototyped a custom print-head shroud in Fusion 360, balancing tight spatial constraints with the integration of upgraded high-performance blower fans.
Execution:	Validated the aerodynamic design intent through rigorous physical stress testing, successfully eliminating thermal inconsistencies across complex overhangs, large-diameter structural rings, and micro-scale precision cones.



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LOGISTICS & AVAILABILITY - JAN — JUL 2027 CO-OP

Availability

Available for a continuous 7-month block, (January 4th, 2027– July 30th, 2027), aligning with standard US corporate Spring/Summer Co-Op cycles. This full-time block is an officially approved, credit-bearing Industrial Attachment, sanctioned by the National University of Singapore (NUS).

Work Authorization — J-1 Intern Visa.

I'm eligible for a J-1 Intern visa, sponsored by Cultural Vistas, a U.S. Department of State–designated Exchange Visitor Program sponsor. This route doesn't require your organization to file an immigration petition — Cultural Vistas manages SEVIS compliance directly.

It does require a Training/Internship Placement Plan (Form DS-7002), completed jointly by the sponsor and host organization and signed by my direct supervisor. I'll draft the plan in full ahead of time — objectives, schedule, and role description — so your team's part is limited to confirming the details and signing.

Tax Note

As a J-1 intern, I'm generally exempt from FICA (Social Security/Medicare) withholding under federal tax rules — a minor payroll simplification, not the basis for this application.

Academic Requirement

NUS requires a midpoint and final evaluation to award course credit for the attachment. I'll pre-fill all technical content and rubric details in advance; your involvement is reviewing and signing.

Relocation

I am fully prepared to independently manage and self-fund my international travel and initial U.S. housing. I will handle my own consular scheduling, embassy appointments, and required insurance, ensuring zero logistical delays for your team prior to day one.

Questions on any of the above — happy to walk through it: **clementtoh@u.nus.edu**
clement.toh2604@gmail.com